In [1]:
```
# # This mounts your Google Drive to the Colab VM.
# from google.colab import drive
# drive.mount('/content/drive')

# # TODO: Enter the foldername in your Drive where you have saved the unzipped
# # assignment folder, e.g. 'cs6353/assignments/assignment2/'
# FOLDERNAME = 'assignment2'
# assert FOLDERNAME is not None, "[!] Enter the foldername."

# # Now that we've mounted your Drive, this ensures that
# # the Python interpreter of the Colab VM can load
# # python files from within it.
# import sys
# sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# # This downloads the CIFAR-10 dataset to your Drive
# # if it doesn't already exist.
# %cd /content/drive/My\ Drive/$FOLDERNAME/cs6353/datasets/
# !bash get_datasets.sh
# %cd /content/drive/My\ Drive/$FOLDERNAME

# # Install requirements from colab_requirements.txt
# # TODO: Please change your path below to the colab_requirements.txt file
# ! python -m pip install -r /content/drive/My\ Drive/$FOLDERNAME/colab_requirement

# --------- Laptop ---------
# %cd "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignment2/cs6353/datasets/"
# !bash get_datasets.sh
# %cd "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assignment2"
# --------- Laptop ---------

# --------- Desktop ---------
%cd "/home/jared/CS-6353/Homeworks/assignment2/cs6353/datasets/"
!bash get_datasets.sh
%cd "/home/jared/CS-6353/Homeworks/assignment2/"
# --------- Desktop ---------

# Install requirements from colab_requirements.txt
# TODO: Please change your path below to the colab_requirements.txt file

# --------- Desktop ---------
! python -m pip install -r "/home/jared/CS-6353/Homeworks/assignment2/colab_require
# --------- Desktop ---------

# --------- Laptop ---------
# ! python -m pip install -r "/home/jared/Desktop/Fall2025/CS-6353/Homeworks/assign
# --------- Laptop ---------
```

```
/home/jared/CS-6353/Homeworks/assignment2/cs6353/datasets
--2025-10-01 19:25:42--  http://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170498071 (163M) [application/x-gzip]
Saving to: 'cifar-10-python.tar.gz'

cifar-10-python.tar 100%[===================>] 162.60M  10.1MB/s    in 17s

2025-10-01 19:25:59 (9.49 MB/s) - 'cifar-10-python.tar.gz' saved [170498071/17049807
1]

cifar-10-batches-py/
cifar-10-batches-py/data_batch_4
cifar-10-batches-py/readme.html
cifar-10-batches-py/test_batch
cifar-10-batches-py/data_batch_3
cifar-10-batches-py/batches.meta
cifar-10-batches-py/data_batch_2
cifar-10-batches-py/data_batch_5
cifar-10-batches-py/data_batch_1
/home/jared/CS-6353/Homeworks/assignment2
Requirement already satisfied: anyio==3.7.1 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 1)) (3.7.1)
Requirement already satisfied: appnope==0.1.3 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 2)) (0.1.3)
Requirement already satisfied: argon2-cffi==23.1.0 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 3)) (23.1.0)
Requirement already satisfied: argon2-cffi-bindings==21.2.0 in /home/jared/anaconda
3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/as
signment2/colab_requirements.txt (line 4)) (21.2.0)
Requirement already satisfied: arrow==1.2.3 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 5)) (1.2.3)
Requirement already satisfied: asttokens==2.2.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 6)) (2.2.1)
Requirement already satisfied: async-lru==2.0.4 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 7)) (2.0.4)
Requirement already satisfied: attrs==23.1.0 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 8)) (23.1.0)
Requirement already satisfied: Babel==2.12.1 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 9)) (2.12.1)
Requirement already satisfied: backcall==0.2.0 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 10)) (0.2.0)
Requirement already satisfied: beautifulsoup4==4.12.2 in /home/jared/anaconda3/envs/
CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignmen
t2/colab_requirements.txt (line 11)) (4.12.2)
```

```
Requirement already satisfied: bleach==6.0.0 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 12)) (6.0.0)
Requirement already satisfied: certifi==2023.7.22 in /home/jared/anaconda3/envs/CS63
53/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/c
olab_requirements.txt (line 13)) (2023.7.22)
Requirement already satisfied: cffi==1.15.1 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 14)) (1.15.1)
Requirement already satisfied: charset-normalizer==3.2.0 in /home/jared/anaconda3/en
vs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assign
ment2/colab_requirements.txt (line 15)) (3.2.0)
Requirement already satisfied: comm==0.1.4 in /home/jared/anaconda3/envs/CS6353/lib/
python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_re
quirements.txt (line 16)) (0.1.4)
Requirement already satisfied: contourpy==1.1.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 17)) (1.1.0)
Requirement already satisfied: cycler==0.11.0 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 18)) (0.11.0)
Requirement already satisfied: debugpy==1.6.7.post1 in /home/jared/anaconda3/envs/CS
6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
2/colab_requirements.txt (line 19)) (1.6.7.post1)
Requirement already satisfied: decorator<=5.0 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 20)) (4.4.2)
Requirement already satisfied: defusedxml==0.7.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 21)) (0.7.1)
Requirement already satisfied: executing==1.2.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 22)) (1.2.0)
Requirement already satisfied: fastjsonschema==2.18.0 in /home/jared/anaconda3/envs/
CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignmen
t2/colab_requirements.txt (line 23)) (2.18.0)
Requirement already satisfied: fonttools==4.42.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 24)) (4.42.1)
Requirement already satisfied: fqdn==1.5.1 in /home/jared/anaconda3/envs/CS6353/lib/
python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_re
quirements.txt (line 25)) (1.5.1)
Requirement already satisfied: idna==3.4 in /home/jared/anaconda3/envs/CS6353/lib/py
thon3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requ
irements.txt (line 26)) (3.4)
Requirement already satisfied: imageio==2.31.1 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 27)) (2.31.1)
Requirement already satisfied: ipykernel<=5.5.6 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 28)) (5.5.6)
Requirement already satisfied: ipython<=7.34.0 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 29)) (7.34.0)
Requirement already satisfied: isoduration==20.11.0 in /home/jared/anaconda3/envs/CS
6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
```

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2/colab_requirements.txt (line 30)) (20.11.0)
Requirement already satisfied: jedi==0.19.0 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 31)) (0.19.0)
Requirement already satisfied: Jinja2==3.1.2 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 32)) (3.1.2)
Requirement already satisfied: json5==0.9.14 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 33)) (0.9.14)
Requirement already satisfied: jsonpointer==2.4 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 34)) (2.4)
Requirement already satisfied: jsonschema==4.19.0 in /home/jared/anaconda3/envs/CS63
53/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/c
olab_requirements.txt (line 35)) (4.19.0)
Requirement already satisfied: jsonschema-specifications==2023.7.1 in /home/jared/an
aconda3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homewo
rks/assignment2/colab_requirements.txt (line 36)) (2023.7.1)
Requirement already satisfied: jupyter-events==0.7.0 in /home/jared/anaconda3/envs/C
S6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
2/colab_requirements.txt (line 37)) (0.7.0)
Requirement already satisfied: jupyter-lsp==2.2.0 in /home/jared/anaconda3/envs/CS63
53/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/c
olab_requirements.txt (line 38)) (2.2.0)
Requirement already satisfied: jupyter_client<8.0 in /home/jared/anaconda3/envs/CS63
53/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/c
olab_requirements.txt (line 39)) (7.4.9)
Requirement already satisfied: jupyter_core==5.3.1 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 40)) (5.3.1)
Requirement already satisfied: jupyter_server==2.7.2 in /home/jared/anaconda3/envs/C
S6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
2/colab_requirements.txt (line 41)) (2.7.2)
Requirement already satisfied: jupyter_server_terminals==0.4.4 in /home/jared/anacon
da3/envs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/
assignment2/colab_requirements.txt (line 42)) (0.4.4)
Requirement already satisfied: jupyterlab==4.0.5 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 43)) (4.0.5)
Requirement already satisfied: jupyterlab-pygments==0.2.2 in /home/jared/anaconda3/e
nvs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assig
nment2/colab_requirements.txt (line 44)) (0.2.2)
Requirement already satisfied: jupyterlab_server==2.24.0 in /home/jared/anaconda3/en
vs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assign
ment2/colab_requirements.txt (line 45)) (2.24.0)
Requirement already satisfied: kiwisolver==1.4.5 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 46)) (1.4.5)
Requirement already satisfied: MarkupSafe==2.1.3 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 47)) (2.1.3)
Requirement already satisfied: matplotlib==3.7.2 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 48)) (3.7.2)
Requirement already satisfied: matplotlib-inline==0.1.6 in /home/jared/anaconda3/env
```

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s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 49)) (0.1.6)
Requirement already satisfied: mistune==3.0.1 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 50)) (3.0.1)
Requirement already satisfied: nbclient==0.8.0 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 51)) (0.8.0)
Requirement already satisfied: nbconvert==7.7.4 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 52)) (7.7.4)
Requirement already satisfied: nbformat==5.9.2 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 53)) (5.9.2)
Requirement already satisfied: nest-asyncio==1.5.7 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 54)) (1.5.7)
Requirement already satisfied: notebook_shim==0.2.3 in /home/jared/anaconda3/envs/CS
6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
2/colab_requirements.txt (line 55)) (0.2.3)
Requirement already satisfied: numpy<1.24,>=1.22 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 56)) (1.23.5)
Requirement already satisfied: overrides==7.4.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 57)) (7.4.0)
Requirement already satisfied: packaging==23.1 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 58)) (23.1)
Requirement already satisfied: pandas<=1.5.3 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 59)) (1.5.3)
Requirement already satisfied: pandocfilters==1.5.0 in /home/jared/anaconda3/envs/CS
6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
2/colab_requirements.txt (line 60)) (1.5.0)
Requirement already satisfied: parso==0.8.3 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 61)) (0.8.3)
Requirement already satisfied: pexpect==4.8.0 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 62)) (4.8.0)
Requirement already satisfied: pickleshare==0.7.5 in /home/jared/anaconda3/envs/CS63
53/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/c
olab_requirements.txt (line 63)) (0.7.5)
Requirement already satisfied: Pillow==10.0.0 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 64)) (10.0.0)
Requirement already satisfied: platformdirs==3.10.0 in /home/jared/anaconda3/envs/CS
6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment
2/colab_requirements.txt (line 65)) (3.10.0)
Requirement already satisfied: prometheus-client==0.17.1 in /home/jared/anaconda3/en
vs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assign
ment2/colab_requirements.txt (line 66)) (0.17.1)
Requirement already satisfied: prompt-toolkit==3.0.39 in /home/jared/anaconda3/envs/
CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignmen
t2/colab_requirements.txt (line 67)) (3.0.39)
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Requirement already satisfied: psutil==5.9.5 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 68)) (5.9.5)
Requirement already satisfied: ptyprocess==0.7.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 69)) (0.7.0)
Requirement already satisfied: pure-eval==0.2.2 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 70)) (0.2.2)
Requirement already satisfied: pycparser==2.21 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 71)) (2.21)
Requirement already satisfied: Pygments==2.16.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 72)) (2.16.1)
Requirement already satisfied: pyparsing==3.0.9 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 73)) (3.0.9)
Requirement already satisfied: python-dateutil==2.8.2 in /home/jared/anaconda3/envs/
CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignmen
t2/colab_requirements.txt (line 74)) (2.8.2)
Requirement already satisfied: python-json-logger==2.0.7 in /home/jared/anaconda3/en
vs/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assign
ment2/colab_requirements.txt (line 75)) (2.0.7)
Requirement already satisfied: pytz==2023.3 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 76)) (2023.3)
Requirement already satisfied: PyYAML==6.0.1 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 77)) (6.0.1)
Requirement already satisfied: pyzmq<25 in /home/jared/anaconda3/envs/CS6353/lib/pyt
hon3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_requi
rements.txt (line 78)) (24.0.1)
Requirement already satisfied: referencing==0.30.2 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 79)) (0.30.2)
Requirement already satisfied: requests==2.31.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 80)) (2.31.0)
Requirement already satisfied: rfc3339-validator==0.1.4 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 81)) (0.1.4)
Requirement already satisfied: rfc3986-validator==0.1.1 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 82)) (0.1.1)
Requirement already satisfied: rpds-py==0.9.2 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 83)) (0.9.2)
Requirement already satisfied: scipy==1.11.2 in /home/jared/anaconda3/envs/CS6353/li
b/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_
requirements.txt (line 84)) (1.11.2)
Requirement already satisfied: seaborn==0.12.2 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 85)) (0.12.2)
Requirement already satisfied: Send2Trash==1.8.2 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
```

```
lab_requirements.txt (line 86)) (1.8.2)
Requirement already satisfied: six==1.16.0 in /home/jared/anaconda3/envs/CS6353/lib/
python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab_re
quirements.txt (line 87)) (1.16.0)
Requirement already satisfied: sniffio==1.3.0 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 88)) (1.3.0)
Requirement already satisfied: soupsieve==2.4.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 89)) (2.4.1)
Requirement already satisfied: stack-data==0.6.2 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 90)) (0.6.2)
Requirement already satisfied: terminado==0.17.1 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 91)) (0.17.1)
Requirement already satisfied: tinycss2==1.2.1 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 92)) (1.2.1)
Requirement already satisfied: tornado<=6.3.2 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 93)) (6.3.2)
Requirement already satisfied: traitlets==5.9.0 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/co
lab_requirements.txt (line 94)) (5.9.0)
Requirement already satisfied: tzdata==2023.3 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 95)) (2023.3)
Requirement already satisfied: uri-template==1.3.0 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 96)) (1.3.0)
Requirement already satisfied: urllib3==2.0.4 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 97)) (2.0.4)
Requirement already satisfied: wcwidth==0.2.6 in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/colab
_requirements.txt (line 98)) (0.2.6)
Requirement already satisfied: webcolors==1.13 in /home/jared/anaconda3/envs/CS6353/
lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/cola
b_requirements.txt (line 99)) (1.13)
Requirement already satisfied: webencodings==0.5.1 in /home/jared/anaconda3/envs/CS6
353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignment2/
colab_requirements.txt (line 100)) (0.5.1)
Requirement already satisfied: websocket-client==1.6.2 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from -r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 101)) (1.6.2)
Requirement already satisfied: exceptiongroup in /home/jared/anaconda3/envs/CS6353/l
ib/python3.10/site-packages (from anyio==3.7.1->-r /home/jared/CS-6353/Homeworks/ass
ignment2/colab_requirements.txt (line 1)) (1.3.0)
Requirement already satisfied: typing-extensions>=4.0.0 in /home/jared/anaconda3/env
s/CS6353/lib/python3.10/site-packages (from async-lru==2.0.4->-r /home/jared/CS-635
3/Homeworks/assignment2/colab_requirements.txt (line 7)) (4.15.0)
Requirement already satisfied: tomli in /home/jared/anaconda3/envs/CS6353/lib/python
3.10/site-packages (from jupyterlab==4.0.5->-r /home/jared/CS-6353/Homeworks/assignm
ent2/colab_requirements.txt (line 43)) (2.2.1)
Requirement already satisfied: ipython-genutils in /home/jared/anaconda3/envs/CS635
```

```
3/lib/python3.10/site-packages (from ipykernel<=5.5.6->-r /home/jared/CS-6353/Homewo
rks/assignment2/colab_requirements.txt (line 28)) (0.2.0)
Requirement already satisfied: setuptools>=18.5 in /home/jared/anaconda3/envs/CS635
3/lib/python3.10/site-packages (from ipython<=7.34.0->-r /home/jared/CS-6353/Homewor
ks/assignment2/colab_requirements.txt (line 29)) (78.1.1)
Requirement already satisfied: entrypoints in /home/jared/anaconda3/envs/CS6353/lib/
python3.10/site-packages (from jupyter_client<8.0->-r /home/jared/CS-6353/Homeworks/
assignment2/colab_requirements.txt (line 39)) (0.4)
```

Implementing a Neural Network

In this exercise we will develop a neural network with fully-connected layers to perform classification, and test it out on the CIFAR-10 dataset.

In [2]:
```
# A bit of setup

from __future__ import print_function
import numpy as np
import matplotlib.pyplot as plt

from cs6353.classifiers.neural_net import TwoLayerNet


%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

# for auto-reloading external modules
# see http://stackoverflow.com/questions/1907993/autoreload-of-modules-in-ipython
%load_ext autoreload
%autoreload 2

def rel_error(x, y):
    """ returns relative error """
    return np.max(np.abs(x - y) / (np.maximum(1e-8, np.abs(x) + np.abs(y))))
```

We will use the class `TwoLayerNet` in the file `cs6353/classifiers/neural_net.py` to represent instances of our network. The network parameters are stored in the instance variable `self.params` where keys are string parameter names and values are numpy arrays. Below, we initialize toy data and a toy model that we will use to develop your implementation.

In [3]:
```
# Create a small net and some toy data to check your implementations.
# Note that we set the random seed for repeatable experiments.

input_size = 4
hidden_size = 10
num_classes = 3
num_inputs = 5
```

```python
def init_toy_model():
    np.random.seed(0)
    return TwoLayerNet(input_size, hidden_size, num_classes, std=1e-1)

def init_toy_data():
    np.random.seed(1)
    X = 10 * np.random.randn(num_inputs, input_size)
    y = np.array([0, 1, 2, 2, 1])
    return X, y

net = init_toy_model()
X, y = init_toy_data()
```

Forward pass: compute scores

Open the file `cs6353/classifiers/neural_net.py` and look at the method `TwoLayerNet.loss` . This function is very similar to the loss functions you have written for the SVM and Softmax exercises: It takes the data and weights and computes the class scores, the loss, and the gradients on the parameters.

Implement the first part of the forward pass which uses the weights and biases to compute the scores for all inputs.

In [4]:
```python
scores = net.loss(X)
print('Your scores:')
print(scores)
print()
print('correct scores:')
correct_scores = np.asarray([
  [-0.81233741, -1.27654624, -0.70335995],
  [-0.17129677, -1.18803311, -0.47310444],
  [-0.51590475, -1.01354314, -0.8504215 ],
  [-0.15419291, -0.48629638, -0.52901952],
  [-0.00618733, -0.12435261, -0.15226949]])
print(correct_scores)
print()

# The difference should be very small. We get < 1e-7
print('Difference between your scores and correct scores:')
print(np.sum(np.abs(scores - correct_scores)))
```

```
Your scores:
[[-0.81233741 -1.27654624 -0.70335995]
 [-0.17129677 -1.18803311 -0.47310444]
 [-0.51590475 -1.01354314 -0.8504215 ]
 [-0.15419291 -0.48629638 -0.52901952]
 [-0.00618733 -0.12435261 -0.15226949]]

correct scores:
[[-0.81233741 -1.27654624 -0.70335995]
 [-0.17129677 -1.18803311 -0.47310444]
 [-0.51590475 -1.01354314 -0.8504215 ]
 [-0.15419291 -0.48629638 -0.52901952]
 [-0.00618733 -0.12435261 -0.15226949]]

Difference between your scores and correct scores:
3.6802720745909845e-08
```

Forward pass: compute loss

In the same function, implement the second part that computes the data and regularization loss.

In [5]:
```
loss, _ = net.loss(X, y, reg=0.05)
correct_loss = 1.30378789133

# should be very small, we get < 1e-12
print('Difference between your loss and correct loss:')
print(np.sum(np.abs(loss - correct_loss)))
```

```
Difference between your loss and correct loss:
1.7985612998927536e-13
```

Backward pass

Implement the rest of the function. This will compute the gradient of the loss with respect to the variables `W1` , `b1` , `W2` , and `b2` . Now that you (hopefully!) have a correctly implemented forward pass, you can debug your backward pass using a numeric gradient check:

In [6]:
```
from cs6353.gradient_check import eval_numerical_gradient

# Use numeric gradient checking to check your implementation of the backward pass.
# If your implementation is correct, the difference between the numeric and
# analytic gradients should be less than 1e-8 for each of W1, W2, b1, and b2.

loss, grads = net.loss(X, y, reg=0.05)

# these should all be less than 1e-8 or so
for param_name in grads:
    f = lambda W: net.loss(X, y, reg=0.05)[0]
```

```
    param_grad_num = eval_numerical_gradient(f, net.params[param_name], verbose=Fal
    print('%s max relative error: %e' % (param_name, rel_error(param_grad_num, grad
```

```
W2 max relative error: 3.440708e-09
b2 max relative error: 4.447656e-11
W1 max relative error: 3.561318e-09
b1 max relative error: 2.738421e-09
```

Train the network

To train the network we will use stochastic gradient descent (SGD), similar to the SVM and Softmax classifiers. Look at the function `TwoLayerNet.train` and fill in the missing sections to implement the training procedure. This should be very similar to the training procedure you used for the SVM and Softmax classifiers. You will also have to implement `TwoLayerNet.predict`, as the training process periodically performs prediction to keep track of accuracy over time while the network trains.

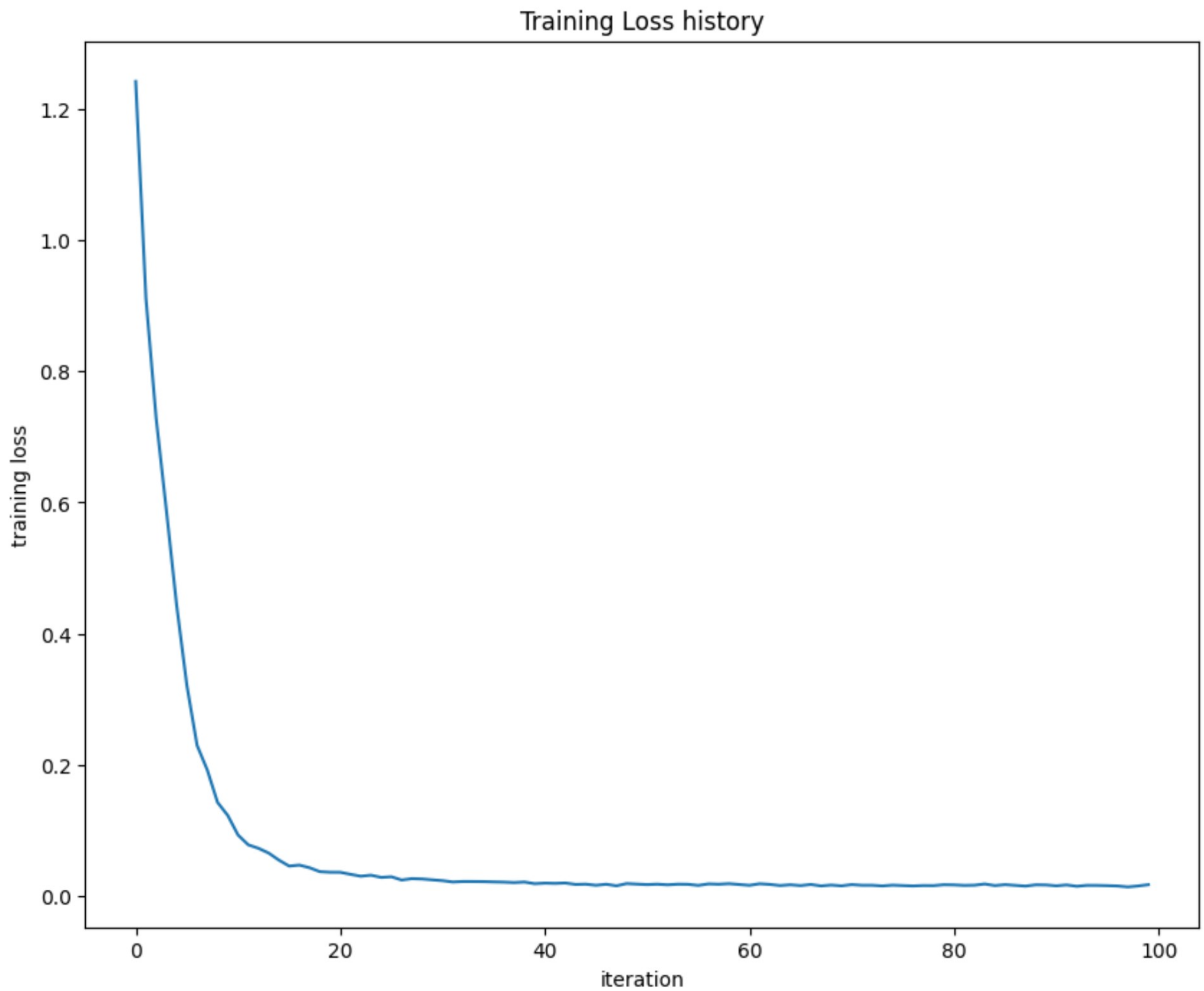
Once you have implemented the method, run the code below to train a two-layer network on toy data. You should achieve a training loss less than 0.2.

In [7]:
```
net = init_toy_model()
stats = net.train(X, y, X, y,
            learning_rate=1e-1, reg=5e-6,
            num_iters=100, verbose=False)

print('Final training loss: ', stats['loss_history'][-1])

# plot the loss history
plt.plot(stats['loss_history'])
plt.xlabel('iteration')
plt.ylabel('training loss')
plt.title('Training Loss history')
plt.show()
```

```
Final training loss:  0.017149607938732O2
```



Load the data

Now that you have implemented a two-layer network that passes gradient checks and works on toy data, it's time to load up our favorite CIFAR-10 data so we can use it to train a classifier on a real dataset.

In [8]:
```
from cs6353.data_utils import load_CIFAR10

def get_CIFAR10_data(num_training=49000, num_validation=1000, num_test=1000):
    """
    Load the CIFAR-10 dataset from disk and perform preprocessing to prepare
    it for the two-layer neural net classifier. These are the same steps as
    we used for the SVM, but condensed to a single function.
    """
    # Load the raw CIFAR-10 data
    cifar10_dir = 'cs6353/datasets/cifar-10-batches-py'

    X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)

    # Subsample the data
    mask = list(range(num_training, num_training + num_validation))
    X_val = X_train[mask]
```

```
    y_val = y_train[mask]
    mask = list(range(num_training))
    X_train = X_train[mask]
    y_train = y_train[mask]
    mask = list(range(num_test))
    X_test = X_test[mask]
    y_test = y_test[mask]

    # Normalize the data: subtract the mean image
    mean_image = np.mean(X_train, axis=0)
    X_train -= mean_image
    X_val -= mean_image
    X_test -= mean_image

    # Reshape data to rows
    X_train = X_train.reshape(num_training, -1)
    X_val = X_val.reshape(num_validation, -1)
    X_test = X_test.reshape(num_test, -1)

    return X_train, y_train, X_val, y_val, X_test, y_test


# Cleaning up variables to prevent loading data multiple times (which may cause mem
try:
   del X_train, y_train
   del X_test, y_test
   print('Clear previously loaded data.')
except:
   pass

# Invoke the above function to get our data.
X_train, y_train, X_val, y_val, X_test, y_test = get_CIFAR10_data()
print('Train data shape: ', X_train.shape)
print('Train labels shape: ', y_train.shape)
print('Validation data shape: ', X_val.shape)
print('Validation labels shape: ', y_val.shape)
print('Test data shape: ', X_test.shape)
print('Test labels shape: ', y_test.shape)
```

```
Train data shape:  (49000, 3072)
Train labels shape:  (49000,)
Validation data shape:  (1000, 3072)
Validation labels shape:  (1000,)
Test data shape:  (1000, 3072)
Test labels shape:  (1000,)
```

Train a network

To train our network we will use SGD. In addition, we will adjust the learning rate with an exponential learning rate schedule as optimization proceeds; after each epoch, we will reduce the learning rate by multiplying it by a decay rate.

In [9]:
```
input_size = 32 * 32 * 3
```

```python
hidden_size = 50
num_classes = 10
net = TwoLayerNet(input_size, hidden_size, num_classes)

# Train the network
stats = net.train(X_train, y_train, X_val, y_val,
            num_iters=1000, batch_size=200,
            learning_rate=1e-4, learning_rate_decay=0.95,
            reg=0.25, verbose=True)

# Predict on the validation set
val_acc = (net.predict(X_val) == y_val).mean()
print('Validation accuracy: ', val_acc)
```

```
iteration 0 / 1000: loss 2.302954
iteration 100 / 1000: loss 2.302550
iteration 200 / 1000: loss 2.297648
iteration 300 / 1000: loss 2.259602
iteration 400 / 1000: loss 2.204170
iteration 500 / 1000: loss 2.118565
iteration 600 / 1000: loss 2.051535
iteration 700 / 1000: loss 1.988466
iteration 800 / 1000: loss 2.006591
iteration 900 / 1000: loss 1.951473
Validation accuracy:  0.287
```

Debug the training

With the default parameters we provided above, you should get a validation accuracy of about 0.29 on the validation set. This isn't very good.

One strategy for getting insight into what's wrong is to plot the loss function and the accuracies on the training and validation sets during optimization.

Another strategy is to visualize the weights that were learned in the first layer of the network. In most neural networks trained on visual data, the first layer weights typically show some visible structure when visualized.

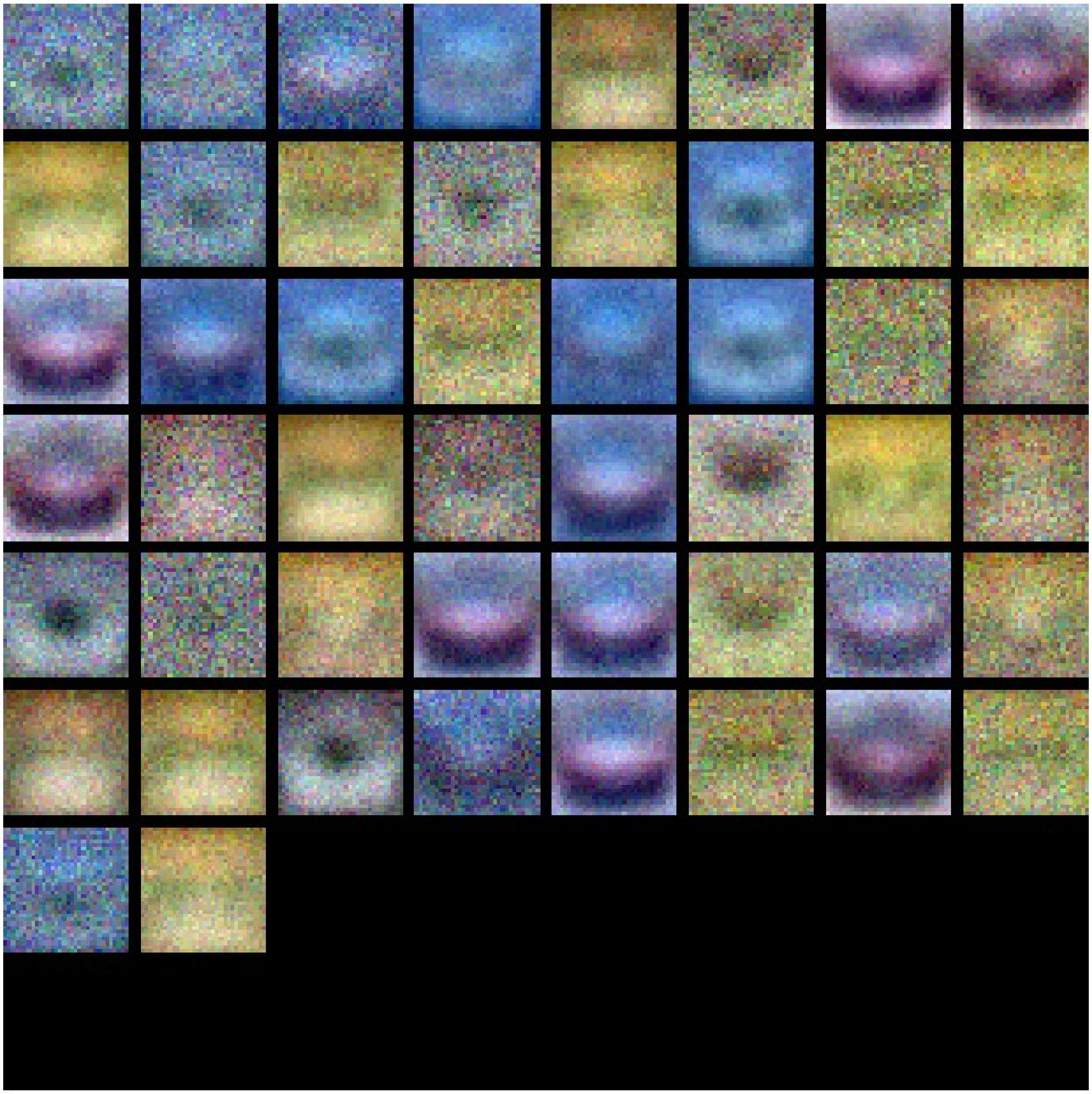
In [10]:

```python
from cs6353.vis_utils import visualize_grid

# Visualize the weights of the network

def show_net_weights(net):
    W1 = net.params['W1']
    W1 = W1.reshape(32, 32, 3, -1).transpose(3, 0, 1, 2)
    plt.imshow(visualize_grid(W1, padding=3).astype('uint8'))
    plt.gca().axis('off')
    plt.show()

show_net_weights(net)
```



Tune your hyperparameters

What's wrong?. Looking at the visualizations above, we see that the loss is decreasing more or less linearly, which seems to suggest that the learning rate may be too low. Moreover, there is no gap between the training and validation accuracy, suggesting that the model we used has low capacity, and that we should increase its size. On the other hand, with a very large model we would expect to see more overfitting, which would manifest itself as a very large gap between the training and validation accuracy.

Tuning. Tuning the hyperparameters and developing intuition for how they affect the final performance is a large part of using Neural Networks, so we want you to get a lot of practice. Below, you should experiment with different values of the various hyperparameters, including hidden layer size, learning rate, number of training epochs, and regularization strength. You might also consider tuning the learning rate decay, but you should be able to get good performance using the default value.

Approximate results. You should be aim to achieve a classification accuracy of greater than 48% on the validation set. Our best network gets over 52% on the validation set.

Experiment: You goal in this exercise is to get as good of a result on CIFAR-10 as you can, with a fully-connected Neural Network. Feel free implement your own techniques (e.g. PCA to reduce dimensionality, or adding dropout, or adding features to the solver, etc.).

In [12]:
```
#################################################################################
# TODO: Tune hyperparameters using the validation set. Store your best trained  #
# model in best_net.                                                            #
#                                                                               #
# To help debug your network, it may help to use visualizations similar to the  #
# ones we used above; these visualizations will have significant qualitative    #
# differences from the ones we saw above for the poorly tuned network.          #
#                                                                               #
# Tweaking hyperparameters by hand can be fun, but you might find it useful to  #
# write code to sweep through possible combinations of hyperparameters          #
# automatically like we did on the previous exercises.                          #
#################################################################################
# Your code
best_net = None # store the best model into this
hidden_layer_sizes = [20, 40, 80, 160]
learning_rates = [1e-3, 1e-2, 1e-1]
num_epochs = [1000, 2000]
regularization_strengths = [0.10, 0.5,1]

best_accuracy = -1
results = {}
best_args = ()

for cur_hidden_layer_size in hidden_layer_sizes:
    for cur_learning_rate in learning_rates:
        for cur_num_epoch in num_epochs:
            for cur_regularization_strength in regularization_strengths:
                input_size = 32 * 32 * 3
                num_classes = 10

                net = TwoLayerNet(input_size, cur_hidden_layer_size, num_classes)

                # Train the network
                stats = net.train(X_train, y_train, X_val, y_val,
                            num_iters=cur_num_epoch, batch_size=200,
                            learning_rate=cur_learning_rate, learning_rate_decay=0.
                            reg=cur_regularization_strength, verbose=False)

                # Predict on the validation set
                val_acc = (net.predict(X_val) == y_val).mean()

                results[(cur_hidden_layer_size, cur_learning_rate, cur_num_epoch, c
                if val_acc > best_accuracy:
                    best_accuracy = val_acc
                    best_args = (cur_hidden_layer_size, cur_learning_rate, cur_num_
```

```
                    best_net = net

# Print out results.
for hidden_layer_size, learning_rate, num_epoch, regularization_strength in results
    cur_val_acc = results[(hidden_layer_size, learning_rate, num_epoch, regularizat
    print('hidden_layer_size: %d learning_rate: %f num_epoch: %d regularization_str
                hidden_layer_size, learning_rate, num_epoch, regularization_strengt

# Print
best_hidden_layer_size, best_learning_rate, best_num_epoch, best_regularization_str
print()
print("The best hyperparameters:")
print('hidden_layer_size: %d learning_rate: %f num_epoch: %d regularization_strengt
            best_hidden_layer_size, best_learning_rate, best_num_epoch, best_regula


# hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_str


################################################################################
#                              END OF YOUR CODE                                #
################################################################################
```

```
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:103: Runt
imeWarning: divide by zero encountered in log
  loss_vec = -1 * np.log(np.exp(correct_class_scores) / np.sum(np.exp(scores), axis
= 1))
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:103: Runt
imeWarning: overflow encountered in exp
  loss_vec = -1 * np.log(np.exp(correct_class_scores) / np.sum(np.exp(scores), axis
= 1))
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:103: Runt
imeWarning: invalid value encountered in divide
  loss_vec = -1 * np.log(np.exp(correct_class_scores) / np.sum(np.exp(scores), axis
= 1))
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:120: Runt
imeWarning: overflow encountered in exp
  s_js = np.exp(scores) / np.sum(np.exp(scores), axis = 1).reshape(-1,1)
/home/jared/CS-6353/Homeworks/assignment2/cs6353/classifiers/neural_net.py:120: Runt
imeWarning: invalid value encountered in divide
  s_js = np.exp(scores) / np.sum(np.exp(scores), axis = 1).reshape(-1,1)
```

```
hidden_layer_size: 20 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.447000
hidden_layer_size: 20 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.455000
hidden_layer_size: 20 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.437000
hidden_layer_size: 20 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.439000
hidden_layer_size: 20 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.455000
hidden_layer_size: 20 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.467000
hidden_layer_size: 20 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 20 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.460000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.461000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.455000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.481000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.500000
hidden_layer_size: 40 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.478000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
```

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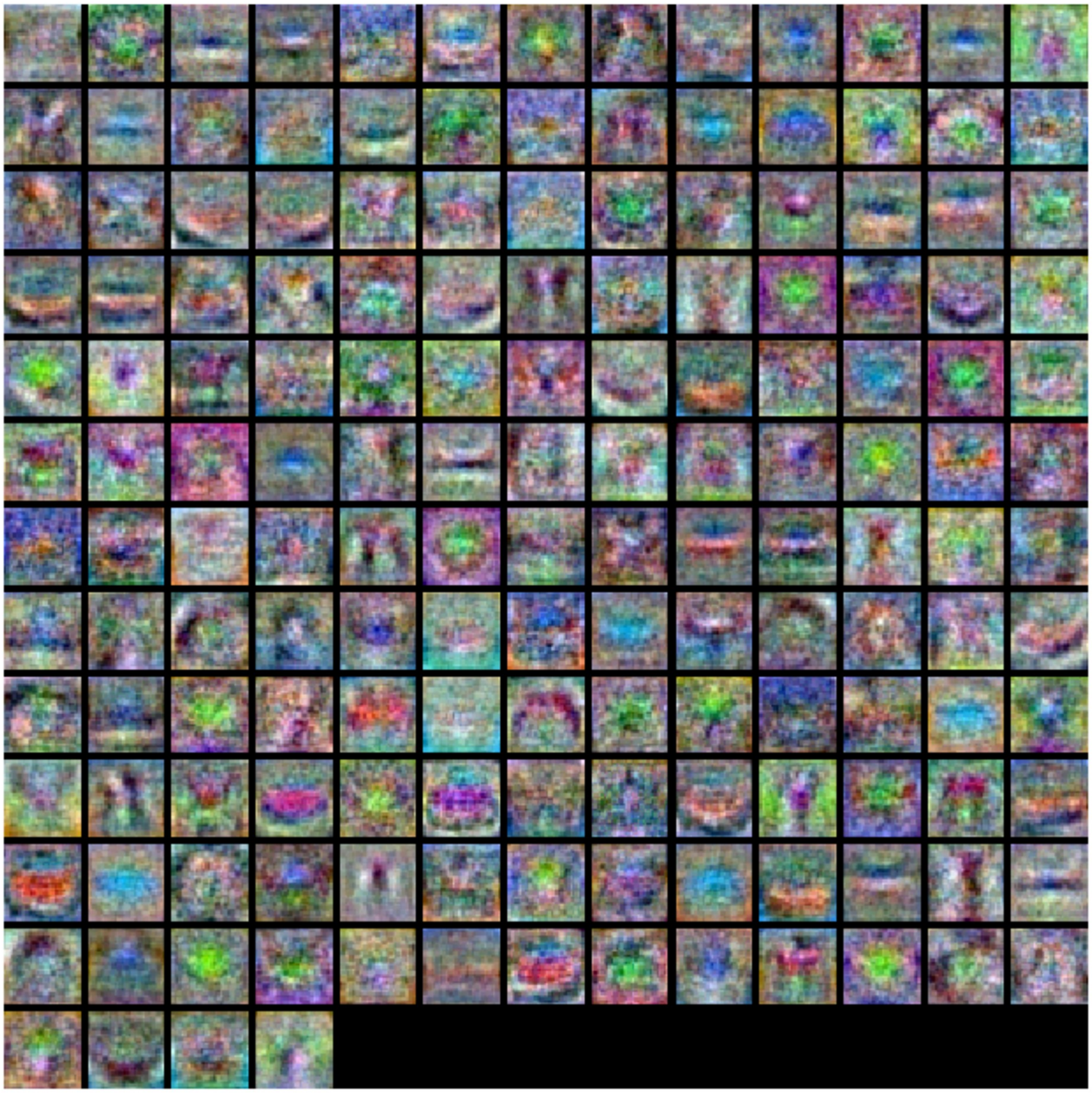
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 40 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.485000
hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.466000
hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.465000
hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.501000
hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.493000
hidden_layer_size: 80 learning_rate: 0.001000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.486000
hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.010000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 1000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 80 learning_rate: 0.100000 num_epoch: 2000 regularization_strengt
h: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 0.100000 validation_accuracy: 0.476000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 0.500000 validation_accuracy: 0.481000
```

```
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 1000 regularization_streng
th: 1.000000 validation_accuracy: 0.476000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 0.100000 validation_accuracy: 0.512000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 0.500000 validation_accuracy: 0.504000
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 1.000000 validation_accuracy: 0.484000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 1000 regularization_streng
th: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 1000 regularization_streng
th: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 1000 regularization_streng
th: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 2000 regularization_streng
th: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 2000 regularization_streng
th: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.010000 num_epoch: 2000 regularization_streng
th: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_streng
th: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_streng
th: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 1000 regularization_streng
th: 1.000000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_streng
th: 0.100000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_streng
th: 0.500000 validation_accuracy: 0.087000
hidden_layer_size: 160 learning_rate: 0.100000 num_epoch: 2000 regularization_streng
th: 1.000000 validation_accuracy: 0.087000

The best hyperparameters:
hidden_layer_size: 160 learning_rate: 0.001000 num_epoch: 2000 regularization_streng
th: 0.100000 validation_accuracy: 0.512000
```

In [13]:

```
# visualize the weights of the best network
show_net_weights(best_net)
```



Run on the test set

When you are done experimenting, you should evaluate your final trained network on the test set; you should get above 48%.

In [14]:
```
test_acc = (best_net.predict(X_test) == y_test).mean()
print('Test accuracy: ', test_acc)
```

```
Test accuracy:  0.529
```

Inline Question

Now that you have trained a Neural Network classifier, you may find that your testing accuracy is much lower than the training accuracy. In what ways can we decrease this gap? Select all that apply.

1. Train on a larger dataset.
2. Add more hidden units.
3. Increase the regularization strength.
4. None of the above.

Your answer:

- Reponse: (1),(3)

Your explanation:

- (1) It is likely that if we train on a larger dataset our model will generalize better. This is dependent on the distribution that the data comes from. If the distribution is the same as the train data but not matching the test data we won't make much progress. If it comes from a distribution that can also our full test set to better align with the true distribution then it could help with generalization and help us decrease the gap.
- (3) Increase the regularization strength. As we have already seen in our class the regularizer is what prevents the model from overfitting to the training data by preferring smaller weights. In this case if we increase our regularization we will get less overfitting to the data and likely better generalizization and test accuracy. This alone will already decrease the gap between our train accuracy and our test accuracy